



ENDOCRINE SYSTEM — FULL MIND MAP

Biology Ch. 7.viii · Human Anatomy & Physiology · Hinglish · SSC / BPSC / BSSC / Railway

📄 Topics in this sub-chapter:

1. Endocrine Glands — Ductless Glands & Pituitary
2. Mixed Gland, Gonads & Basic Hormone–Gland Matching
3. ADH & Urine Control
4. Pancreas, Islets of Langerhans & Insulin
5. Secretin & Gastrin — Digestive Hormones
6. Oxytocin — Mother–Baby Bonding, Milk Ejection & Misuse
7. Parathyroid Gland & Calcium–Phosphate Regulation
8. Thyroid, Iodine, Goitre, Thyroxine & TSH
9. Adrenaline — Fight or Flight
10. Estrogen, Sex Hormones, Menopause & Other Hormonal Facts
11. Melatonin & Circadian Rhythm
12. Pheromones & Important Matching Questions

1. Endocrine Glands — Ductless Glands & Pituitary

- Endocrine glands ductless glands hoti hain—inke excretory ducts nahi hote aur hormones directly blood mein release hote hain. Major endocrine glands mein pineal, pituitary, thymus, thyroid, parathyroid aur adrenal glands include kiye gaye hain. [Q1]
- Pituitary brain ke base par hypothalamus ke neeche sella turcica naam ke bony hollow mein located hoti hai. Source ise “Master gland” aur hypophysis cerebri bhi kehta hai. [Q2, Q3]

⚠️ EXAM TRAP — Q2: “Smallest Endocrine Gland”

- ▶ Q2 ka marked answer given options mein Pituitary (D) hai. Usi source explanation ki last line overall smallest endocrine gland Pineal (~7.5 mm) batati hai. Dono source statements preserve kiye gaye hain. [Q2]

2. Mixed Gland, Gonads & Basic Hormone–Gland Matching

- Pancreas mixed gland hai—endocrine role mein insulin, glucagon, somatostatin aur pancreatic polypeptide se blood sugar regulate karta hai; exocrine role mein pancreatic juice pancreatic duct se duodenum mein release karta hai. [Q4]
- Corpus luteum female ovary mein temporary endocrine structure hai aur relatively high progesterone ke saath moderate estradiol aur inhibin A produce karta hai. [Q5]
- Leydig/interstitial cells testes ke seminiferous tubules ke paas hoti hain aur mainly androgen, especially testosterone, secrete karti hain. [Q6]
- Important gland→hormone match: Gonads→Progesterone, Pituitary→Growth hormone, Pancreas→Insulin, Adrenal→Cortisol. Hormone→gland match: Progesterone→Ovaries, Testosterone→Testes, Thyroxine→Thyroid, Insulin→Pancreas. [Q7, Q9]

3. ADH & Urine Control

- ADH (Antidiuretic Hormone) source ke according pituitary se released hota hai aur kidney se urine ki quantity/secretion ko control karta hai. [Q8]

4. Pancreas, Islets of Langerhans & Insulin

- Carbohydrate-rich diet ke baad bhi blood glucose relatively constant rehne ka source reason pancreatic hormone insulin hai; beta cells insulin se blood glucose maintain karti hain. Insulin deficiency diabetes aur excess insulin hypoglycemia se linked hai. [Q10]
- Pancreas defective hone par insulin aur glucagon formation badly affect hota hai. Source: beta cells→insulin, alpha cells→glucagon, delta cells→somatostatin. [Q11]

- Insulin protein-chain/peptide hormone hai, source ke according 51 amino acids ka micropeptide; beta cells of Islets of Langerhans ise produce karti hain. [Q13, Q14, Q16, Q17, Q18, Q19, Q20, Q21]
- Insulin carbohydrates, fats aur proteins ke metabolism ko regulate karta hai by promoting glucose absorption into liver, fat aur skeletal muscle cells. [Q15]
- Source insulin ke chemical properties mein metal ion Zinc mention karta hai. [Q22]
- Source Q23 ke explanation mein insulin ke sources ke roop mein “roots of dahlias” aur pancreatic beta cells dono mention kiye gaye hain; is source wording ko as-is preserve kiya gaya hai. [Q23]

⚠ **EXAM TRAPS — Insulin Source-Specific Points**

- ▶ Q15 source ka scientifically broader explanation carbohydrates + fats + proteins teenon metabolism batata hai, isliye page par “More than one” answer hai; note mein BPSC final key ne option (B) diya tha, jise source “not completely appropriate” kehta hai. [Q15]
- ▶ Q23 source explicitly “roots of dahlias” ko insulin source ke roop mein mention karta hai. Is unusual source wording ko bina outside correction ke preserve kiya gaya hai. [Q23]

5. Secretin & Gastrin — Digestive Hormones

- Secretin pancreas ko digestive juice production ke liye stimulate karta hai; source ise duodenum ki S cells ka peptide hormone batata hai aur duodenal pH/stomach-liver secretions se bhi link karta hai. [Q12]
- Hormone–function pair mein “Gastrin→regulation of blood pressure” incorrect pair hai. Source gastrin ko gastric cells se gastric acid secretion stimulate karne wala peptide hormone batata hai. [Q24]

6. Oxytocin — Mother–Baby Bonding, Milk Ejection & Misuse

- Mother ka baby ko hug/kiss karna oxytocin secretion se linked hai; source ise “love/cuddle hormone” kehta hai aur childbirth, maternal bonding aur breast-feeding se associate karta hai. [Q25]
- Oxytocin uterus aur breast par act karta hai; labour ke during cervix/uterus stimulation aur mammary glands se milk release/ejection mein role play karta hai. [Q26]
- Source oxytocin ke misuse ko cattle mein milk production badhane aur pumpkins/watermelons jaise fruits/vegetables ka size badhane se link karta hai; labour speed-up misuse bhi mention hai. [Q27]
- Source Q28 mein keratin, fibroin aur collagen ko proteins batata hai, jabki oxytocin ko peptide hormone/neuropeptide batakar “not a full structural protein” framing use karta hai. [Q28]
- Cows/buffaloes mein udders tak milk descend karwane ke liye source oxytocin/Pitocin injection batata hai. [Q29]
- Milk-ejection hormone oxytocin source question ke hisab se pituitary gland se secreted/released maana gaya hai; explanation ke mutabik ye hypothalamic nuclei mein produced aur posterior pituitary se released hota hai. [Q30]

⚠ **EXAM TRAP — Q28 Oxytocin Classification**

- ▶ Q28 ka source answer Oxytocin ko “not a protein” select karta hai, while explanation ise peptide hormone/neuropeptide bolkar structural proteins (keratin/fibroin/collagen) se distinguish karti hai. [Q28]

7. Parathyroid Gland & Calcium–Phosphate Regulation

- Parathyroid glands pituitary gland se independent function kar sakti hain. Source parathyroid hormone aur calcitonin ko blood/bone calcium regulation se link karta hai. [Q31]
- Parathyroid hormone (PTH) chief cells se secreted polypeptide hai aur blood calcium regulate karta hai; source PTH ko proximal renal tubule mein phosphate reabsorption reduce karke urine phosphate excretion badhane se bhi link karta hai. [Q49]

8. Thyroid, Iodine, Goitre, Thyroxine & TSH

- Source ke according iodised salt useful hai because it “controls the thyroid gland”; iodine deficiency thyroid problems se linked hai. [Q32]
- Goitre iodine deficiency se linked thyroid enlargement hai; source hilly areas mein low iodine in soil/water/agricultural produce ko endemic goitre ka explanation deta hai. [Q33]
- Thyroxine thyroid gland ka main hormone hai; heart/digestive function, metabolism, brain development, bone health aur muscle control se linked hai. [Q34]
- Thyroxine (T4) aur triiodothyronine (T3) iodine-containing hormones hain; source ke mutabik thyroid cells iodine + amino acid tyrosine se inhe banati hain. T4 mein 4 iodine atoms aur T3 mein 3 hote hain. [Q35, Q36, Q37]
- Thyroid hormone output ko TSH regulate karta hai; TSH anterior pituitary se produced hota hai aur thyroid ko thyroxine secrete karne ke liye stimulate karta hai. [Q38]

9. Adrenaline — Fight or Flight

- Important match: Adrenaline→Anger/fear/danger, Estrogen→Females, Insulin→Glucose, Pheromones→partner attraction through smell. [Q39]
- Adrenaline/Epinephrine “fight or flight hormone” hai, adrenal glands se released hota hai. Source increased blood supply/activity, cardiac muscle contraction, heartbeat/pulse/arterial pressure/cardiac output increase aur excitement feeling describe karta hai. [Q40, Q41]

10. Estrogen, Sex Hormones, Menopause & Other Hormonal Facts

- Estrogen primarily ovaries ke follicles se produced hota hai; corpus luteum aur placenta bhi secrete karte hain. Source female secondary sexual characteristics, endometrium aur menstrual cycle se link karta hai. [Q42, Q43, Q44]
- Source Estrone ko weak postmenopausal estrogen aur minor female sex hormone batata hai; estrogen/oestrogen actual female sex hormone ke roop mein explained hai. [Q45]
- Menopause source ke according ovarian estrogen aur progesterone secretion ke end se marked hai; ovaries small testosterone aur adrenal gland androstenedione continue kar sakte hain, jo body fat mein estrogen forms mein convert ho sakte hain. [Q46]
- Somatotropin/Growth hormone exercise ke saath increase hota hai. Source progesterone ko small amount mein male testes mein aur testosterone ko small amount mein women ke ovaries/adrenal glands mein bhi produced batata hai; stress cortisol release ko increase karta hai. [Q47]

⚠ SOURCE NOTE — Q46 Cancelled by UPPSC

- ▶ Q46 ke source explanation ke mutabik initial UPPSC key mein option (C) mark hua tha, lekin revised answer key mein question cancel kar diya gaya. Isliye final correct option source mein intentionally blank hai. [Q46]

11. Melatonin & Circadian Rhythm

- Melatonin pineal gland ka hormone hai aur circadian rhythm regulate karne mein help karta hai. Source ke according darkness mein melatonin production badhta hai aur bright light mein ghatta hai. [Q50]

12. Pheromones & Important Matching Questions

- Quick biomolecule match: Hormone→Testosterone, Enzyme→Lipase, Phospholipid→Lecithin, Polymer→Polythene. [Q48]
- Pheromones environment mein release hone wali chemical substances hain jo same species ke dusre organisms ke behaviour/physiology ko affect karti hain; source especially insects se associate karta hai. [Q51, Q52]